

Conclusively identified as *Saturnia pavonia*!

Classification result (scientific name, common name, confidence %)

1. <i>Saturnia pavonia</i> (Small Emperor Moth)	89.4%
2. <i>Saturnia pavoniella</i> (Ligurian Emperor)	3.3%

► Show thinking (5 steps)

Saturnia pavonia — small emperor moth

Saturnia pavonia is the small emperor moth, usually called simply the emperor moth in Britain. It is the only member of the Saturniidae family found in the British Isles.

- **Male:** about 60 mm wingspan, brown and white with orange-red markings and prominent eyespots; it flies by day, mainly from mid-April to late June.
- **Female:** larger—about 80 mm—and generally greyer and less brightly coloured; it is usually active at night.
- **Habitat:** especially heathland and moorland.
- **Caterpillar:** initially black and orange, later green with black rings and yellow and red markings. Heather is its most commonly recorded food plant, though it uses many other plants.
- **Life cycle:** it spends the winter as a pupa in a fibrous cocoon.

Status in the UK

The available field-guide material confirms it as a British species but does **not** provide a precise UK rarity category, population assessment, or specific legal designation.

The dedicated status check found:

- no explicit UK legal-protection status in the retrieved material;
- no explicit IUCN category;
- no entry classifying it as an invasive species in the UK.

Those are limits of the available information, rather than proof that no other legislation or local recording rules could apply.

What you should do

If you have found one, the appropriate response is normally simply to **leave it undisturbed**. Do not use pesticide or remove it just because it is present. You can photograph it from a distance and note the date, location, habitat and—if it is a caterpillar—the plant it was feeding on.

The supplied material does not report a toxin or sting associated with *S. pavonia*, but avoiding unnecessary handling of moths and caterpillars is still a sensible precaution.

🚫 **This may be a regionally invasive species** — check local reporting/disposal rules before doing anything with it.

Sources:

- Britain's Butterflies, Westwood, J. O. (John Obadiah), 1805-1893, 1887 (WILDGuides field guide, modern; reference/citation only.) — <https://archive.org/details/b28052328>
- British Butterflies, 1890 (W. S. Coleman, Routledge, 1860. Corrected title (was invented as A History of British Butterflies).) — <https://archive.org/details/historyofbritish03morr>
- British Entomology (John Curtis), Curtis, John, 1791-1862, 1824 (Illustrated early UK insect reference.) — <https://archive.org/details/britishentomolog01curt>
- The Malay Archipelago, Wallace, Alfred Russel, 1869 (Alfred Russel Wallace.) — <https://archive.org/details/malayarchipelago02wall>
- Wikipedia, "Cinnabar moth" — https://en.wikipedia.org/wiki/Cinnabar_moth
- Wikipedia, "Coccinella septempunctata" — https://en.wikipedia.org/wiki/Coccinella_septempunctata
- Wikipedia, "Cydalima perspectalis" — https://en.wikipedia.org/wiki/Cydalima_perspectalis
- Wikipedia, "Green-veined white" — https://en.wikipedia.org/wiki/Green-veined_white
- Wikipedia, "Marsh fritillary" — https://en.wikipedia.org/wiki/Marsh_fritillary
- Wikipedia, "Oak processionary" — https://en.wikipedia.org/wiki/Oak_processionary
- Wikipedia, "Pieris rapae" — https://en.wikipedia.org/wiki/Pieris_rapae
- Wikipedia, "Saturnia pavonia" — https://en.wikipedia.org/wiki/Saturnia_pavonia
- Wikipedia, "Saturnia pavoniella" — https://en.wikipedia.org/wiki/Saturnia_pavoniella

What is the scientific name of the small emperor moth and who first described it?

► Show thinking (2 steps)

The small emperor moth's scientific name is *Saturnia pavonia*. It was first described by **Carl Linnaeus in 1758**, so the taxonomic citation is *Saturnia pavonia* (**Linnaeus, 1758**).

Sources:

- Britain's Butterflies, Westwood, J. O. (John Obadiah), 1805-1893, 1887 (WILDGuides field guide, modern; reference/citation only.) — <https://archive.org/details/b28052328>
- British Butterflies, 1890 (W. S. Coleman, Routledge, 1860. Corrected title (was invented as A History of British Butterflies).) — <https://archive.org/details/historyofbritish03morr>
- British Entomology (John Curtis), Curtis, John, 1791-1862, 1824 (Illustrated early UK insect reference.) — <https://archive.org/details/britishentomolog01curt>
- Wikipedia, "Saturnia pavonia" — https://en.wikipedia.org/wiki/Saturnia_pavonia
- Wikipedia, "Saturnia pavoniella" — https://en.wikipedia.org/wiki/Saturnia_pavoniella

What family does it belong to and where can it be found?

► Show thinking (2 steps)

The small emperor moth belongs to the **family Saturniidae**.

It is found throughout the **Palearctic region**, including the **British Isles**, where it is the only member of the Saturniidae family. It is most often associated with **heathland and moorland**.

Sources:

- British Butterflies, 1890 (W. S. Coleman, Routledge, 1860. Corrected title (was invented as A History of British Butterflies).) — <https://archive.org/details/historyofbritish03morr>
- British Entomology (John Curtis), Curtis, John, 1791-1862, 1824 (Illustrated early UK insect reference.) — <https://archive.org/details/britishentomolog01curt>
- Wikipedia, "Aconitum" — <https://en.wikipedia.org/wiki/Aconitum>
- Wikipedia, "Green-veined white" — https://en.wikipedia.org/wiki/Green-veined_white
- Wikipedia, "Oak processionary" — https://en.wikipedia.org/wiki/Oak_processionary
- Wikipedia, "Pieris rapae" — https://en.wikipedia.org/wiki/Pieris_rapae
- Wikipedia, "Saturnia pavonia" — https://en.wikipedia.org/wiki/Saturnia_pavonia
- Wikipedia, "Saturnia pavoniella" — https://en.wikipedia.org/wiki/Saturnia_pavoniella

What are the main differences in appearance between males and females?

► Show thinking (2 steps)

- **Male:** Smaller, with a wingspan of about **60 mm**; generally **brown and white**.
- **Female:** Larger, reaching about **80 mm**, and typically **greyer** in overall colour.

Both sexes have the moth's conspicuous **eyespots and similar wing pattern**, so size and ground colour are the easiest visual differences.

Sources:

- Britain's Butterflies, Westwood, J. O. (John Obadiah), 1805-1893, 1887 (WILDGuides field guide, modern; reference/citation only.) — <https://archive.org/details/b28052328>
- British Entomology (John Curtis), Curtis, John, 1791-1862, 1824 (Illustrated early UK insect reference.) — <https://archive.org/details/britishentomolog01curt>

When do males fly and what do they look for?

► Show thinking (2 steps)

Male small emperor moths fly **rapidly during the day**, mainly from **mid-April to late June**. They are searching for the relatively sluggish **females**, which usually fly at night.

Sources:

- Britain's Butterflies, Westwood, J. O. (John Obadiah), 1805-1893, 1887 (WILDGuides field guide, modern; reference/citation only.) — <https://archive.org/details/b28052328>
- Wikipedia, "Coccinella septempunctata" — https://en.wikipedia.org/wiki/Coccinella_septempunctata
- Wikipedia, "Green shield bug" — https://en.wikipedia.org/wiki/Green_shield_bug
- Wikipedia, "Green-veined white" — https://en.wikipedia.org/wiki/Green-veined_white
- Wikipedia, "House sparrow" — https://en.wikipedia.org/wiki/House_sparrow
- Wikipedia, "Pieris rapae" — https://en.wikipedia.org/wiki/Pieris_rapae
- Wikipedia, "Saturnia pavonia" — https://en.wikipedia.org/wiki/Saturnia_pavonia
- Wikipedia, "Saturnia pavoniella" — https://en.wikipedia.org/wiki/Saturnia_pavoniella
- Wikipedia, "Speckled wood (butterfly)" — https://en.wikipedia.org/wiki/Speckled_wood_%28butterfly%29

What habitats does this moth prefer?

► Show thinking (2 steps)

The small emperor moth (*Saturnia pavonia*) uses a range of habitats but is most strongly associated with **heathland and moorland**. Its caterpillars commonly feed on heather in these habitats.

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- Britain's Butterflies, Westwood, J. O. (John Obadiah), 1805-1893, 1887 (WILDGuides field guide, modern; reference/citation only.) — <https://archive.org/details/b28052328>
- British Entomology (John Curtis), Curtis, John, 1791-1862, 1824 (Illustrated early UK insect reference.) — <https://archive.org/details/britishentomolog01curt>
- Wikipedia, "Aconitum" — <https://en.wikipedia.org/wiki/Aconitum>
- Wikipedia, "Marsh fritillary" — https://en.wikipedia.org/wiki/Marsh_fritillary
- Wikipedia, "Meadow brown" — https://en.wikipedia.org/wiki/Meadow_brown
- Wikipedia, "Saturnia pavonia" — https://en.wikipedia.org/wiki/Saturnia_pavonia
- Wikipedia, "Saturnia pavoniella" — https://en.wikipedia.org/wiki/Saturnia_pavoniella

What does the caterpillar look like and what does it eat?

► Show thinking (2 steps)

The caterpillar changes appearance as it grows: it is **black and orange at first**, later becoming **green with black rings and yellow and red spots**.

It eats a wide range of plants, but **heather is its most common food plant**. Other recorded food plants include **heaths, birch, alder, hawthorn, oak, willow, bramble, rose, apple, and meadowsweet**.

Sources:

- Britain's Butterflies, Westwood, J. O. (John Obadiah), 1805-1893, 1887 (WILDGuides field guide, modern; reference/citation only.) — <https://archive.org/details/b28052328>
- British Butterflies, 1890 (W. S. Coleman, Routledge, 1860. Corrected title (was invented as A History of British Butterflies).) — <https://archive.org/details/historyofbritish03morr>
- Wikipedia, "Cinnabar moth" — https://en.wikipedia.org/wiki/Cinnabar_moth
- Wikipedia, "Saturnia pavonia" — https://en.wikipedia.org/wiki/Saturnia_pavonia
- Wikipedia, "Saturnia pavoniella" — https://en.wikipedia.org/wiki/Saturnia_pavoniella

What is a common incorrect genus name sometimes used for this species?

► Show thinking (2 steps)

The incorrect genus name sometimes used is ***Pavonia***. The accepted name is ***Saturnia pavonia***.

Sources:

- Britain's Butterflies, Westwood, J. O. (John Obadiah), 1805-1893, 1887 (WILDGuides field guide, modern; reference/citation only.) — <https://archive.org/details/b28052328>
- British Entomology (John Curtis), Curtis, John, 1791-1862, 1824 (Illustrated early UK insect reference.) — <https://archive.org/details/britishentomolog01curt>
- Wikipedia, "Saturnia pavonia" — https://en.wikipedia.org/wiki/Saturnia_pavonia
- Wikipedia, "Saturnia pavoniella" — https://en.wikipedia.org/wiki/Saturnia_pavoniella

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